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Determining emissions using AIS

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1. Introduction

The aim of Work Package 4 (WP 4) is to investigate whether real-time measurement data of ship positions, so-called Automatic Identification System (AIS) can be used 1) to improve the quality and internal comparability of existing statistics and 2) to produce new statistical products relevant to the European Statistical System (ESS). AIS is a mandatory collision avoidance system on ships. Each ship transmits a signal providing details of the ship's speed and location at frequent intervals. AIS data are picked up at numerous ground stations at ports and on the coast by various agencies globally. They are also picked up by satellites.

Five National Statistical Institutes participate in WP4: the national statistical institutes of The Netherlands (Work package leader), Denmark[1], Greece, Norway and Poland. WP4 is subdivided into two phases. Specific Grant Agreement 1 (SGA-1) focused on creating a common database, linking AIS-data to maritime statistics and constructing sea traffic analyses (February 2016-July 2017, See Annex 1 for a more detailed description). The current SGA-2 focuses on the calculation of emissions and future perspectives for AIS data as source data for new statistical output (August 2017-May 2018, see Annex 2 for a more detailed description). At the start of SGA-2 we had to change the scope of SGA-2, because good models for calculating emissions already existed and we could not get access to AIS data from the European Maritime Safety Agency (EMSA). Annex 3 describes the scope of SGA-2 after these changes.

The current SGA will result in three new deliverables, of which this deliverable is the first one:

- Deliverable 4.6: "Deriving emissions using AIS" (to be delivered March 31st 2018)
- Deliverable 4.7: "Possibilities for new statistical output based on European AIS data" (to be delivered March 31st 2018)
- Deliverable 4.8: Consolidated report on project results (to be delivered May 31st 2018)

Besides these new deliverables, we also complemented Deliverable 4.3 during SGA-2 with an extra chapter on the "Quality of satellite data from LuxSpace" (delivered January 31st 2018)

In this Deliverable we investigate how emissions from maritime ships can be determined using AIS data. Shipping impacts the environment in various ways, ranging from wildlife collisions, sound pollution, and atmospheric pollution. Exhaust gases result in both conventional pollutants and greenhouse gas emissions. In the Kyoto Protocol (1997) countries have committed themselves to reduce their anthropogenic greenhouse gas emissions. To maintain this policy, it is necessary to monitor chemical pollutants such as Carbon Dioxide($C0_2$), Methane (CH4), Nitrous Oxide, Nitrogen Dioxide (NO_X), Sulphur Dioxide (SO_2,) and Particulate Matter (PM). AIS provides a method to determine the route of a ship. Combined with a model to estimate the emission of vessels (which depends on factors like travel distance, speed, draught, weather conditions and characteristics of the vessel itself), emissions of e.g. CO_2 and NO_X can be estimated per ship and per (national) territory.

To identify the goals for this WP, we begin by investigating the needs of stakeholders/ experts on this topic in chapter 2. Chapter 3 describes various papers and how existing methodology would be useful

[1] Only in SGA-1, they withdraw from SGA-2

for our purposes. In Chapter 4 recommendations are provided as to how stakeholders' needs on a national and European level can be met.

2. Stakeholders' needs

There are two approaches to determine and publish on emissions: the territorial principle or the residence principle. In the territorial approach emissions are calculated for confined geographical areas, such as a country or a sea. In the residence approach emissions are calculated for the whole world but only for ships that are owned by companies from a certain nationality. This is the approach necessary for the National Account System. Here, we investigated the needs of stakeholders with regards to information on emissions from AIS. First, we looked at how data on emissions are reported internationally and then we focus on national agencies from the countries in this WP.

International organisations

Data on greenhouse gas emissions are officially reported under the United Nations Framework Convention on Climate Change (UNFCCC). Data on air pollutants are officially reported under the Convention on Long-Range Transboundary Air Pollution (CLRTAP) of the EMEP Programme (Co-operative Programme for Monitoring and Evaluation of the Long-Range Transmission of Air pollutants in Europe). The European Environment Agency (EEA) collects national inventories for greenhouse gases and other air pollutants; compiles the EU aggregates; and provides environmental information to policymakers and the public. The EEA works in close collaboration with the European Environmental Information and Observation Network (Eionet), which is a partnership network of 33 member countries. Eurostat's database republishes the most relevant data from these inventories[2].

There are two internationally established approaches to reporting air emissions (emissions of greenhouse gases and air pollutants):

1. The above mentioned national inventories for greenhouse gases and other air pollutants compiled and published by the European Environment Agency (EEA);
2. Air emissions accounts (AEA) produced by Eurostat under Regulation (EU) No 691/2011 on European environmental economic accounts.

The main differences between the two are:

National inventories for greenhouse gases and other air pollutants (territorial principle)	Air emissions accounts (residence principle)
Emissions are assigned to the country where the emission takes place.	Emissions are assigned to the country where the economic operator causing the emission is resident.
Emissions are assigned to processes classified according to their technical nature (e.g. combustion in power plants, solvent use).	Emissions are classified by economic activity, following the NACE classification of the system of national accounts.
Emissions from international navigation and aviation are assigned	Emissions from international navigation and aviation are assigned

[2] Source: http://ec.europa.eu/eurostat/web/environment/air-emissions-inventories

to the countries where the associated fuel is bunkered, irrespective of the operator's place of residence.	to the countries where the operator of the ship/aircraft is resident, regardless of where the emission takes place.

National and EU totals differ between the two approaches, as different boundaries apply.

Eurostat does not report on emissions per se. nevertheless, some publications report on emissions. At the moment Eurostat's estimations of emissions from transport are calculated via fuel consumption, i.e. fuel consumption is multiplied by a specific emission factor. Emissions from maritime transport are treated separately as it is extraterritorial (like international aviation) according to the territorial principle (following the Kyoto protocol). For the air emissions accounts approach, emissions from international navigation should be assigned to the country where the operator of the ship is resident (where the owner is not always the operator), regardless of where the emission takes place. The problem lies in assigning the air emissions following the residence principle.

National offices

Besides the Eurostat (re-)publications, some national statistics are published on emissions. These national statistics are partly created as an input for the UNFCCC reports mentioned above.

Greece

The Ministry of Environment and Energy (MEEN) is the governmental body responsible for the provision of information concerning the state of the environment in Greece in compliance with relevant requirements defined in international conventions, protocols and agreements. According to the Law 3017/2002, the MEEN is responsible for the compilation of the annual report on climate change air emissions ("*Annual Inventory Submission under the Convention and the Kyoto Protocol for Greenhouse and other gases*"). The National Technical University of Athens (NTUA) / School of Chemical Engineering has the technical and scientific responsibility of this inventory for all sectors except the land use, land-use change and forestry (LULUCF) sector. This data is transmitted to the United Nations Framework Convention on Climate Change (UNFCCC) Secretariat and CLRTAP (EEA) and is the official data of Greece on air emissions.

The environmental economic accounts (National Accounts Matrix Environmental Accounts –Air, NAMEA) are compiled by the Section of Construction and Environment Statistics of ELSTAT (the Hellenic Statistical Authority) pursuant to Regulation (EU) No 691/2011. The main data source of NAMEA is the MEEN inventory mentioned above. ELSTAT works in close cooperation with the team responsible for the compilation of the inventory. The allocation of air-emission to the sectors of economic activities is done by the MEEN. ELSTAT works in close cooperation with the technical and scientific team responsible for the compilation of the inventory.

ELSTAT's National Account department and the Environment Section support the effort to investigate the use of AIS to estimate ships' emissions. However, they have not expressed any specific needs.

The Netherlands

The two approaches on reporting emissions are conducted by different institutions. Within Statistics Netherlands, the department of National Accounts reports on the emissions per sector and the Pollutant Release and Transfer Register (PRTR, or so-called "Emission Registration") is a separate governmental

organisation that also reports on the territorial aspects of emissions to national and international organisations such as the EEA[8].

The National Accounts division currently reports on the emissions from the maritime sector by using the financial developmental of Dutch maritime companies in terms of profits as a proxy for the development of emissions. The data is reported to Eurostat. In the end, of course they would be very interested in getting the actual emissions for all Dutch owned ships. However, as a first step the National Accounts division would already be improved by having access to year-to-year development of distance travelled by Dutch owned ships. Then, instead of using profits as a proxy for the development of emissions, the distance travelled would be the proxy for the development of emissions. For emissions by inland ships this distance travelled is already used.

Different organizations collaborate in the PRTR to collect and formally establish release of pollution for the Dutch territory and the continental shelf. The Maritime research Institute Netherlands (MARIN) performs emission calculations on the basis of certain key figures delivered by PRTR and uses AIS data from the Dutch coast guard. As for inland waterways, PRTR is interested in having access to the speed of the ships on Dutch locations (preferably 100 meter intervals) to start off.

Norway

Statistics Norway produces various emission statistics:

- Emissions from business units that are resident in Norway
  - Emissions from Norwegian economic activity: https://www.ssb.no/en/natur-og-miljo/statistikker/nrmiljo
- Emissions from Norwegian territory (for sea traffic, emissions are included when they stem from traffic between two Norwegian harbours)
  - Emissions of greenhouse gases: https://www.ssb.no/en/natur-og-miljo/statistikker/klimagassn
  - Emissions of acidifying gases and ozone precursors: https://www.ssb.no/en/natur-og-miljo/statistikker/agassn

The emissions from sea traffic are calculated from estimated fuel consumption using emission factors. The emission factors are very good. In the case of CO_2 the factors are almost exact. In the case of NO_X, updated factors using information from the NO_X-fund (https://www.nho.no/Prosjekter-og-programmer/NOx-fondet/The-NOx-fund/) are used. The most problematic part is the fuel consumption estimate. Precise information about fuel bought within Norway is available, but fuel from abroad is problematic. Therefore, there is potential for improving the emission statistics by including AIS-based fuel or emission estimates.

Poland

Environmental economic accounts are satellite accounts with respect to the national accounts, i.e. they provide data completely consistent with the national accounts. They provide an important source of information to make environmental decisions. They contribute to assess trends in the use of natural resources, emissions as a result of business activities or an economic activity aimed at the environmental protection.

According to Regulation of the European Parliament and of the Council No 691/2011 of 6 July 2011 on European environmental economic accounts, those accounts cover the three following modules:

1. Air emissions accounts : the emission of six greenhouse gases (including CO_2, and a biomass CO_2 used as fuel) and seven air pollutants into the atmosphere, broken down by 64 industries that cause those emissions, and households;

2. Environmentally related taxes broken down by economic activity: environmental taxes and charges for four generally defined groups: energy, transportation, pollution and resources, broken down by 64 industries that pay those taxes and charges, and households;

3. Economy-wide flow accounts: the value of material inputs, accumulation into the economy, and material outflows into other economies or, back to the environment.

The air emissions accounts record and present data on air emissions in a way that is compatible with the system of national accounts. They record national economies' emissions to air in a breakdown by economic activities, production and consumption, according to ESA3 (European System of National and Regional Accounts). Those accounts record flows of gases and particulate materials originating from the national economy and flowing into the atmosphere. The data are compiled according to NACE Rev.2, the industry standard classification system used in the European Union. The volume of emissions from particular sources and of types (as defined by the law) may be established as a result of measurements or calculated on the basis of fuel-resource balance using pollution indicators specific for technological processes. Both sources are used.

Data on the volume of greenhouse gases emitted in transportation are provided by the National Centre for Emission Balancing and Management - Environmental Protection Research Institute[3].

The department of Emission Accounts is also interested in using AIS data. They are interested in all ships but especially interested in ships that are owned by Polish companies.

3. Literature review

To investigate what work has been done on AIS and emissions on a national level (Norway and the Netherlands) and on a more global level, we will discuss a few papers here. We started with papers focusing on two geographical areas of the members of this WP, then going into a paper widening the geographical area of interest. The first paper by Hulskotte (2014) analyses part of a canal in the Netherlands for inland ships only. Next, the paper by Mjelde, Martinsen, Eide and Endresen (2014) focuses on the seas around Norway and the Arctic Sea. Finally, in the Ship Traffic Emission Assessment 3 Model (STEAM3 model) by Johansson, Jalkanen, and Kukkonen(2017) emissions on all world seas are analysed. For these papers we will discuss the scope (e.g. type of ships, geographical area), the assumptions of the model and give an overview of input and output variables. We will finish by discussing which model is most suitable for our current European purposes. We will not discuss the actual emission factors used, as these are not relevant per se to the present goal of investigating the use of AIS to calculate emissions.

[3] Environment 2017; http://stat.gov.pl/en/topics/environment-energy/environment/environment-2017,1,9.html

Inland shipping in the Netherlands

Scope

The research presented by Hulskotte (2014) calculates emissions of NO_X by inland ship types for a small area in the Amsterdam-Rijnkanaal in the Netherlands by using AIS. Emission calculations are based on speed of the ship and propulsion power estimates for the level of loading. Propulsion power estimation is based on the so-called EMS-model (Emissionregistration and Monitoring Shipping, Bolt, 2003) and then AIS data is used to calculate some variables.

Assumptions of the model

For each ship, propulsion power curves are calculated for different speeds based on the assumption of a ship being empty, fully loaded or partly (2/3) loaded. The level of loading (payload and weight of goods) is defined as a function of the draught of the ship. The average speed of a ship is also calculated. The actual speed is then compared to the ship's average speed. If the speed is 0.3 knots faster than average speed, the ship is assumed to be empty. If the speed is 0.3 knots slower, the ship is assumed to be fully loaded. If it is between the 0.3 boundaries, the ship is assumed to be partly loaded. To calculate emissions, the corresponding propulsion power for the actual speed and level of lading is selected and multiplied by the emission factor. Propulsion power estimates are thus based on ship's length, width, draught and speed over water.

$$P_b = 2U_{rel}R_{tot}$$

where:

P_b is the propulsion power at the brake,

U_{rel} is the speed over water and

R_{tot} is the total Resistance

(Bolt, 2003).

If there is a water flow that is either opposite or similar to the ship's direction it will of course have an impact on the propulsion power that has to be delivered to maintain a certain speed. Here, speed over ground (SOG) is used as a substitute as water speed is assumed to be zero. SOG is derived from the AIS signal. As the other ship variables are usually not reliable, these are based on other sources. The dimensions of the ships are based on a list compiled from websites containing ship information (e.g. marinetrafic.com). The draught of the ships in a (un)loaded condition is based on a list compiled on the basis of certain classes of ships (Hulskotte, 2012).

Waterway dimensions needed for the EMS-model are based on information from the Dutch waterway authority.

The main assumption is speed being dependent on level of loading (draught), other factors such as wind, speed and direction of water flow and traffic are not taken into account. These factors might have also influenced the propulsion power needed. Preliminary analyses for Rijkswaterstaat show that speeds by a ship on a certain trajectory do not show much variation. One could conclude from this that the degree of loading does not vary much, but this could also mean that the 0.3 knots speed difference assumption of

loading is too coarse. Furthermore, the method works for a specific region, but does not calculate the complete journey of a ship. Also, the list of draught of ships in different conditions would have to be expanded greatly to have world coverage.

Input variables

- Propulsion power estimation:
  - ship's length
  - width
  - depth
  - speed over water (here, SOG is used)
- Water dimensions:
  - width on the surface
  - depth
  - transversal section
  - flow rate
  - flow direction
- Emission factor FOR NO_X

Output variables

NO_X is calculated for the total confined geographical area.

Norwegian approach

Scope

The work done by Mjelde, Martinsen, Eide and Endresen (2014) calculates emissions of $C0_2$, NO_X, SO_2, C0 and PM for the seas surrounding Norway (Norwegian economic zone) and the Arctic Sea. The seas are divided into areas in different ways. This is done for all ship types, which are categorized in 13 ship types (oil tankers, chemical / product tankers, gas tankers, bulk ships, general cargo ships, container ships, ro-ro cargo, reefer ships, passenger ships, offshore supply ships, other offshore service vessels, other activities, fishing vessels.

Information based on AIS is available for the main geographical areas:

- The sea around the Norwegian economic zone (see www.havbase.no for the area and emissions statistics)
- Artic Sea: www.havbase.no/havbase_arktis (emissions statistics not included)

Statistics are calculated as aggregates by combining: sea areas, time periods, ship types and pollution types.

Assumptions of the model

- Emission per pollution type = emission factor x fuel consumption
- Fuel consumption for a given engine = installed power (kW) x SFOC (specific fuel consumption in kg fuel/kW h) x load factor x time (h)
- Load factor of main engine = (speed/ service speed)^{3}based on the Propeller law. Service speeds is the average speed under normal load and weather conditions.
- Average speed over 6 minutes, calculated using the great circle distance between the two successive AIS positions.
- Load factor of auxiliary engine depend on ship operation mode.

- Ship operation mode (open sea or harbour mode) is based on the ship's speed over a period of time.
- Calculated for the given position each 6-minute for each ship (time = 0.1 h)

Fuel quality is a complicating factor in the case of SO_2 because ships can use different fuel qualities in different areas as ordered by different emission requirements. Furthermore, the model does not include load and weather conditions, instead long term averages are assumed. The model works satisfactory for vessels travelling from port to port, but is problematic for other vessels such as fishery ships. Fishery ships and other ships being at the same place for a longer period are consuming fuel in a way that cannot be estimated from AIS. The motors may be used in order to remain at the same location and motors can be used to do "fishery work" and other kinds of work.

Input variables

- AIS position for each 6-minute interval
- Fuel specific emission factor for each pollution type ($C0_2$, NO_X, SO_2, C0, PM)
- Ship specific constants:
    - Service speed
    - Engine specific constants (main and auxiliary engine)
        - Installed power (kW)
        - SFOC (kg fuel/kWh)

Output variables

Output for each 6-minute for each ship:

- Tons of each pollution type ($C0_2$, NO_X, SO_2, C0, PM)
- Distance

This is stored together with time, location at the end of the 6 minute interval, ship type and other meta information.

STEAM3 model

Scope

The STEAM3 model can be used in any area with AIS data available (Johansson, Jalkanen, and Kukkonen, 2017). It is used for all seagoing ships worldwide, see the IHS classification of ships (see Annex 4).

Assumptions of the model

The STEAM3 model is based on Finnish methodology. It determines emissions from a ship's route from AIS and a database with detailed technical data of the engines. It also takes into account the influence of emission control areas, scrubbers and dual fuels. The emissions are then computed based on the relationship of the instantaneous speed to the design speed.

A route generation algorithm was developed to take care of missing geographical (AIS) information. This algorithm does not simply perform a linear interpolation, but takes into account the plausibility of a route with regard to geography. For example, in the case of a linear interpolation crossing of land mass, a more realistic route is calculated.

Technical information on the ship itself (e.g., main engine stroke type and rpm) is based on a combination of IHS Fairplay data, an internal ship database and other online sources. When there is no IMO-information, a most similar vessel algorithm is selected by combining different information sources such as AIS and online sources (using a web crawler) and comparing this to the internal database.

The predictions of the emissions of SO_X and CO_2 are based on engine specific fuel consumption.If the data is missing, the average values for that specific ship type are used.

Propelling power is determined by wave height and direction. Hourly significant wave heights and wave direction data are obtained from the WAve Model (WAM); the grid resolution of WAM is supplemented with winds from the HIRLAM model (High Resolution Limited Area Model) which provides numerical weather prediction information, and the expertise associated with it. Additional power requirement depends on parameters describing the wet surface and the three-dimensional structure of the hull (these are different, for example, for oil tankers and passenger ships), and the contact angle between the hull and waves). Main and auxiliary engine power levels are multiplied by emission factors. These are based on measured emissions levels and installed abatement techniques or on the IMO NO_X curve (IMO, 1973):

Emission factor (g/kWh)= 8<17, for engines _ 130 rpm (engine's rated speed)
45.0 _ n−0.2, for engines130<n<2000, n= engine rpm
9.8, for engines over 2000 rpm

SOx emission is calculated based on fuel S-%. The emission factor of the total PM is calculated as a function of fuel sulphur content (mass-based percentage), based on the data from the second IMO Greenhouse Gas study (Buhaug et al., 2009). The lines in Figure 3.1 indicate the PM component emission factors in STEAM; the emission factors of the total PM, SO_4 and H_2O are linearly dependent on the fuel sulphur content.

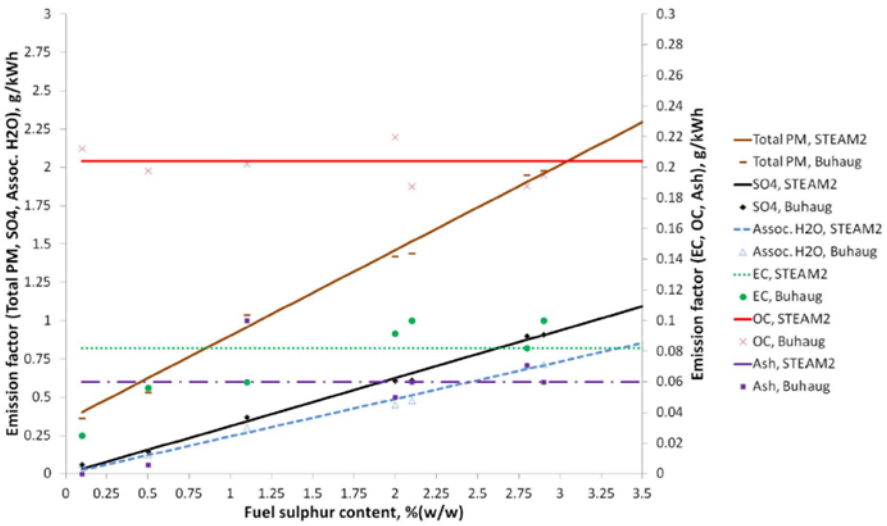

Figure 3.1 Extension of an assessment model for PM and CO1[4]

The STEAM3 model does not currently contain factors for the effects of: sea currents, sea ice, shallow water phenomena (e.g., squat) and hull fouling. Catamaran and trimaran structures or hydrofoil designs are not modeled

Input variables

STEAM3 is based on two main parts of data:

- Automatic Identification System (AIS)
- Ship database with detailed technical information on ships(IHS Fairplay, ship owners and local maritime authorities) containing:
    - engines, generators
    - auxiliary engines
    - physical dimensions, hull form
    - measured emission factors
    - emission abatement techniques installed
    - year of build
    - propelling power (number of propellers, diameter, rpm, CPP/FP)
    - fuel type, sulphur content, specific consumption
    - engine load fuel consumption/emissions, power transmission
    - capacity, reefer containers, cabins

Output variables

On a minute-to-minute basis:

- Tons of each pollutant type (NO_x,SO_x, , CO_2 , CO and PM: Elementary Carbon (EC), Organic Carbon (OC), Ash, Sulphate (SO_4,)

Emissions by:

- Ship type
- Ship age
- Engine stroke type
- Flag state of a ship (based on country code in the MMSI, Maritime Mobile Service Identity)
- Geographical area (if covered by AIS)
- Different time intervals

Conclusions

The models presented here all build on the same principle of using speeds compared to maximum speeds as a proxy for propulsion power used. The methods applied become more advanced in the STEAM3 model. The first and second method are not sufficient for a European analysis because they only cover smaller geographical area. For a larger scale analysis two aspects are needed that are included in the STEAM3 model. The first is an algorithm dealing with missing AIS data due to scarcity of AIS in some areas. Second, is a method to obtain detailed vessel information. To this end, an extended/automatic means of

[4] Source: ''Extension of an assessment model of ship traffic exhaust emissions for particulate matter and carbon monoxide''

combining multiple sources is needed. Finally, the STEAM3 model also takes into account control areas, scrubbers and dual fuels.

Therefore, we conclude that the STEAM3 model would best serve the purpose of analysing emissions of maritime shipping in large geographical areas or even worldwide shipping. However it also shows that local models do quite well for their own area and are much simpler. Start off with the local models might thus be a good first step, before turning to the more complex global model (of course also depending on the needs).

4. Recommendations for European statistics on emissions

All in all, the comments we received from stakeholders were mainly geared towards the National Accounts approach. We hardly received wishes on the territorial approach. This might be because it was harder for us to contact the institutions that work on this. We cannot conclude on the basis the investigation we did whether there is interest in work on the territorial approach.

Prerequisites for emission calculations

When constructing emission statistics on maritime (and inland waterway) transport, a number of data sets are indispensable:

- Worldwide AIS data

  As one of the two of the main approaches, National Account data concerns data from ships owned by companies with a specific nationality. Territory is irrelevant. This means that worldwide data on for example Greek-owned ships is necessary.

- Route generation algorithm

  As some geographical information from AIS is missing and these cannot always be constructed by a simple linear interpolation, a more complex route generation algorithm has to be applied. This algorithm has to take into account geographical information. As such an algorithm already exists, the possibility to obtain this from other parties has to be examined.

- IHS Fairplay registry

  Many specific ship variables are necessary, about e.g. (auxiliary) engines, physical dimensions propelling power. These are for some part available in the IHS Fairplay registry, but the registry is does not completely cover all ships and all aspects of the ships in the registry.

- Web scraping techniques

  Since IHS Fairplay does not fully cover all ship aspects, part of the missing information can be recovered from web sites on ships.

- Similar vessel algorithm

  As the ship registry constructed from HIS Fairplay and web scraping will still not be fully covering all ships, an algorithm is needed that calculates the most similar vessel and uses information from that vessel to complete shop information on ships missing this information in available sources. For the route generation, such an algorithm already exists. Again, the possibility to obtain this from other parties has to be examined.

- Emission factors

  We have not looked at specific emission factors in the present study. However, different factors are already used, so research to be done on which ones would be optimal in terms of reliability and maintenance.

- Emission calculation algorithm

  Like the other algorithms mentioned, the possibility to obtain an emission calculation algorithm from other parties has to be examined.

- Storage

All size of data has to be gained and stored, necessitating a robust way of data storage.

Quick wins before emission calculations

Although the STEAM3 model provides us with a method to determine actual emissions per ship and region, it takes a big investment in terms of getting data, algorithms and storage. Before actually calculating emissions for a ships' journey and aggregating this to either the residency level or territorial level, an intermediate step might already be possible. Stakeholders have expressed interest in two specific aspects that might be somewhat easier to obtain. Firstly, the Dutch department of National Accounts has expressed interest in getting the development of the distance travelled for Dutch-owned ships. Even for European AIS data would be an enrichment of the current method. Secondly, the Dutch PRTR would be interested in getting information on average speeds over 100 meters for Dutch territory. Although to start with, the interest here is mainly on inland waterway ships.

Efficiency gains and other opportunities

To reduce the amount of AIS data to be analysed, a frame of ships could be constructed that enter European waters. This filters out ships that would not be relevant for European statistics. Of course, it has to be checked if there are ships owned by European companies that do not enter European waters. For the residency approach, the frame of ships would only have to contain ships owned by European companies.

To further limit the amount of data to be processed, "average" paths could be used. Journeys between specific ports often follow a similar path. This depends on a number of factors, such as type of ship, time of the year and regulations. For efficiency gain, these "average" paths could be constructed (Marine Traffic has already worked on this) and could be used to limit calculations.

The STEAM3 model does currently not incorporate variables like sea currents, sea ice, or temporary obstructions. These factors have an effect on the propelling power needed or the route to be taken. Research can look into other sources for this.

Besides these efforts to calculate emissions per ship's journey, a possibility to simplify calculations of emissions would be to calculate emissions on a more aggregated level. This could be done using average speeds for ships in a certain area. A first step on a tool for facilitating this, was developed in this WP: a speed grid.

Speed grid

The average speeds in European waters can be calculated in the same manner as the traffic intensities that were calculated in Deliverable 4.3. Basically, the average Speed Over Ground (SOG) in the AIS data is used in the code for calculating intensities. However, there was one methodological issue that had to be solved for calculating the average speed. When calculating the average speed along a trajectory, the average speed should be calculated using the harmonic mean. Then, the average speed in a certain area should be calculated using the arithmetic mean. Therefore, Europe was subdivided in the same cells as for the intensity grid. For each ship, the harmonic mean of the speed was calculated along the trajectory within the cell. After that, the arithmetic mean was calculated over all ships within each grid cell, rendering the average speed per cell.

This is visualized in Shiny in combination with Leaflet. Shiny is a web application framework for programming language R, with which one can create interactive web applications. Leaflet is a JavaScript library for interactive maps, which can also be used from R. The code can be found in https://github.com/mputs/WP4/speed.

Figure 4.1 shows a snap shot of this visualization of the speed grid. In the visualization, a day can be selected from a date list in the bottom. A slider is available for selecting a saturation threshold for the visualization. With this, all cells where the average speed is higher than the threshold can be displayed in intense red (see for example the South-east coast of Portugal) and all cells with a lower speed compared to the threshold as less red (see for example the North Sea and East Sea, which is a special Emission Control Zone as defined by the International Convention for the Prevention of Pollution from Ships, 1973 (MARPOL Protocol). Zooming in on the visualization, the lower average speed in the Dutch inland waterways can also be seen. If regions are not displayed in red, there is no data available on that specific day (in this case the Dirkzwager data).

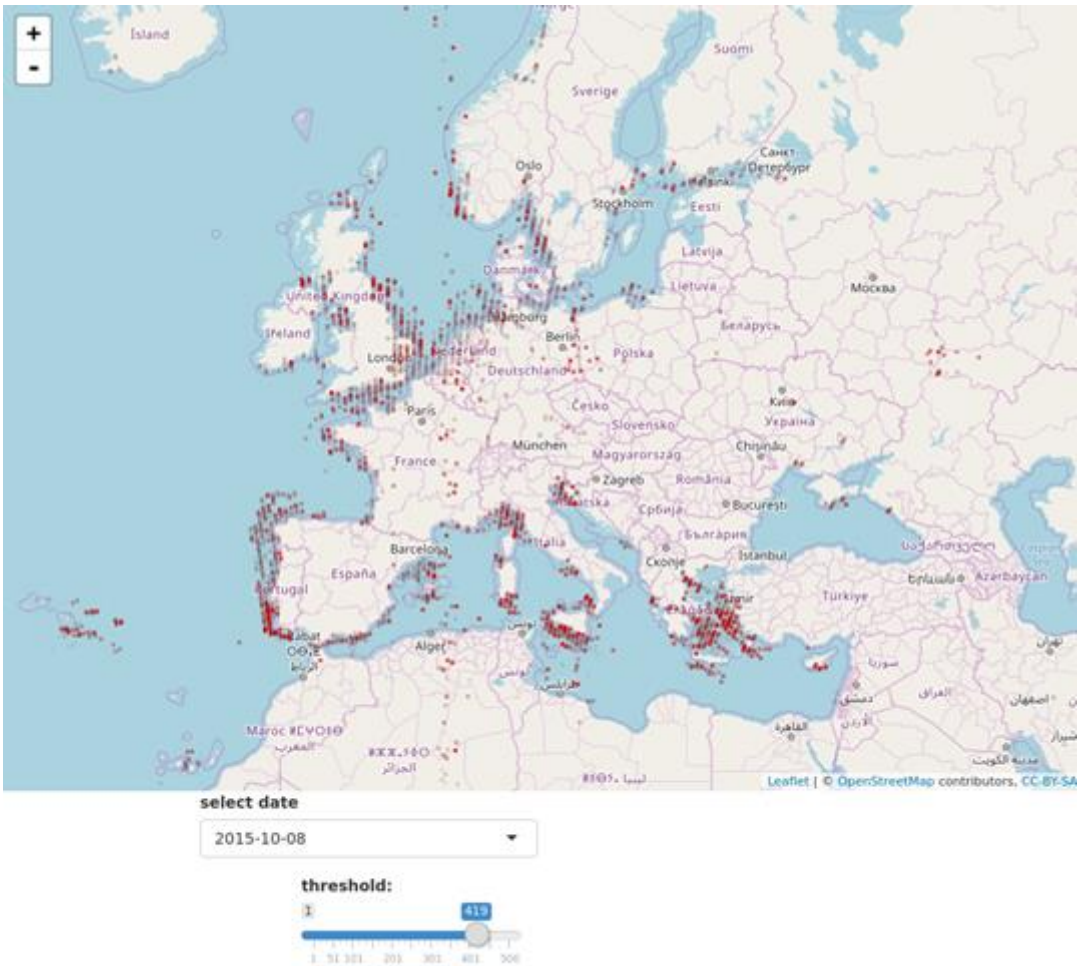

Figure 4.1 Speed grid of Europe

The slider thus gives insight into whether areas have a lower or higher average speed compared to a certain threshold. As speed of a ship is one of the important factors for calculating the emissions, the

speed grid can give a quick glimpse on which areas probably suffer from higher emissions. Another important factor in the level of emissions in a certain area is the number of ships in that certain area. Therefore, we will also work on a merge of the visualizations on speed grid and traffic intensities (to follow).

This visualization could also provide insight into the consequences of regulations that may limit the maximum speed in some areas. But a speed grid based on the maximum speed travelled could gave even more insight the consequences of regulations (see also Deliverable 4.7).

We would like to adjust this algorithm by calculating the speed of ships based on the real distances and time travelled and compare the calculated speeds with the speeds available in the AIS data.

Positioning

The analysis of AIS data for statistics eminently lends itself to process on a European level. Many ships enter multiple European ports and waters. Having every country determine the route of this ship would result in redundant work. Processing of all different sources is quite data intensive. Also, the territorial approach is only half of the approaches of interest, as the National Accounts are interested in worldwide emissions. Therefore, to optimize efficiency on a European level, it would be advisable to perform calculations per (European-owned) ship and then aggregate this either to nationality of owner or to territory a ship traverses. Gaining access to worldwide AIS data & IHS registry on a collaborative European level is also more cost efficient than every country having to buy these data sets separately.

5. References

1. Bolt E., Schatting energiegebruik van binnenvaartschepenversie 3, 22 oktober 2003
2. Hulskotte, J., and Bolt, E. (2012). EMS-protocol Emissies door Binnenvaart: Verbrandingsmotoren.
3. Hulskotte, J. (2014) Emissieberekening binnenvaart op basis van AIS-gegevens. TNO Rapport: TNO 2015 R10067
4. Jalkanen, J.-P., Brink, A., Kalli, J., Pettersson, H., Kukkonen, J., and Stipa, T. (2009). A modelling system for the exhaust emissions of marine traffic and its application in the Baltic Sea area, Atmos. Chem. Phys., 9, 9209-9223, https://doi.org/10.5194/acp-9-9209-2009.
5. Jalkanen, J.-P., Johansson, L., Kukkonen, J., Brink, A., Kalli, J., and Stipa, T. (2012). Extension of an assessment model of ship traffic exhaust emissions for particulate matter and carbon monoxide, Atmos. Chem. Phys., 12, 2641-2659, https://doi.org/10.5194/acp-12-2641-2012.
6. J.-P., Johansson, L. Modelling ship emissions (presentation)
7. J.-P., Johansson, L. Ship emission modelling (presentation)
8. Johansson, L., Jalkanen, J.-P., Kukkonen, J. (2017). Global assessment of shipping emissions in 2015 on a high spatial and temporal resolution. Atmospheric Environment, 167, 403-415, DOI: 10.1016/j.atmosenv.2017.08.042
9. Mjelde, A, Martinsen, K. , Eide, M., and Endresen, Ø. Environmental accounting for Arctic shipping – A framework building on ship tracking data from satellites, In Marine Pollution Bulletin, Volume 87, Issues 1–2, 2014, Pages 22-28, ISSN 0025-326X, https://doi.org/10.1016/j.marpolbul.2014.07.013
10. Information on the Pollutant Release and Transfer Register www.emissieregstratie.nl/erpubliek/content/explanation.en.aspx
11. Additional information is obtained from contact with the first author of the paper and people at Kystverket – The Norwegian Coastal Administration (www.kystverket.no)
12. Web sites in Norwegian: www.havbase.no and www.havbase.no/havbase_arktis

Annex 1: SGA-1 of WP4 in more detail

Work package number	4	Start date: End date:			1.2.2016 31.7.2017		
Title	**AIS Data**						
Partner/co-beneficiary	**NL** **120**	DK 60	EL 40	NO 67	PL 40		

Aim of this work package is to investigate whether real-time measurement data of ship positions (measured by the so-called AIS-system) can be used 1) to improve the quality and internal comparability of existing statistics and 2) for new statistical products relevant for the ESS. Improvement of quality and internal comparability can be obtained e.g. by developing a reference frame of ships and their travels in European waters and then linking this reference frame, by ship number, to register-based data about marine transport from port authorities. These linked data can then be used for emission calculations. New products can be developed for e.g. traffic analyses. The added value of running a pilot with AIS-data at European level is that the source data are generic word wide and data can be obtained at European level. Challenges ahead with this dataset are: obtaining the data at European level, processing and collecting the data in such way that they can be used for multiple purposes, and visualising the results. A part of this work package is also to look into AIS analyses done by others and to investigate the possibility of obtaining already processed data as input for creating comparable official statistics. Especially it is important to make contact with other public authorities. This work package may require data acquisition in collaboration with Eurostat.

Methodological, quality and technical results of the work package, including intermediate findings, will be used as inputs for the envisaged WP 8 of SGA-2, in case SGA-2 will be realised. When carrying out the tasks listed below, care will be taken that these results will be stored for later use, by using the facilities described at WP 9.

Task 1 – Data access.

AIS-data are available for national territories and the entire European territory. For example, AIS-data in the Netherlands are provided by Rijkswaterstaat (a government agency which is part of the Ministry of Infrastructure and the Environment). It is expected that similar agencies in other countries have the AIS-data for their national territories. At the European level, a dataset of AIS data is available at the European Maritime Safety Agency (EMSA). The advantage of using one AIS-dataset for the entire European territory is a) a better comparison of international traffic between the countries and b) more synergy as all participating countries work on the same dataset. A disadvantage is that these data are stored by private companies and handling fees have to be paid. Aim of this task is to decide how European data could be

used for this project, to investigate the possibilities of acquiring data from EMSA (to be coordinated with Eurostat) and, if European data are too costly or too hard to obtain, how national datasets can be obtained.

This task will involve:

- Exploration of the possibilities to collect the data at a European (or worldwide) level.

Participants: NL, DK, EL, NO, PL

Task 2 – Data handling

Aim of this task is to process and store the data in such a way that they can be used for consistent multiple outputs, like

- linking AIS-data with data from port authorities
- traffic analyses
- Inference of journeys from AIS data.

Key elements of this task are:

- which programming language and environment should be used for transformation
- where will the data be processed (in each NSI, by NSIs at European level, by data holders)
- how can we create an environment which is easily accessible for all partners

Participants: NL, NO, PL, and possibly DK (if national sources are used)

Task 3 – Methodology and Techniques

Develop traffic statistics: Linking with data from port authorities.

AIS-data may be linked to data from port authorities. Added value of linking AIS-data to data from port authorities is that the same reference population (= ship number) is used in all harbours. As the journeys and harbour visits of ships can be derived from AIS this linking provides the ESS information about the origin/destination of the cargo, too.

Aims of this task are:

- to build a reference frame of ships in European water (based on AIS-data)
- to find out how data from port authorities can be linked to AIS-data
- to check whether information improves the quality of current statistical outputs and provides more information about the origin/destination of the cargo.

Participants: NL, NO, PL

Traffic analyses

The number of ships during a certain time interval at certain coordinates (like inland waterways or at certain points at sea) can be calculated by AIS-data. This possibility will be explored because this information could be interesting for traffic analyses and economic analyses.

Aims of this task are:

- calculate the number of ships at certain coordinates
- visualise the results to analyse variations in time

Participants: NL, NO, PL, EL

Estimate emissions (envisaged under SGA-2)

This task involves following individual vessels through time. Consequently we can infer the journeys from the data. Combined with a model to estimate the emission of vessels (which depends on travel distance, speed, draught, weather conditions and characteristics of the vessel itself), emissions of e.g. CO_2 and NO_x can be estimated per ship and per national territory.

An advantage of doing these analyses on a European scale– instead of the national level – is that more precise estimates for emissions at national territories can be made.

Aim of this task is 1) to infer journeys from AIS-data, 2) visualise the results, 3) combine these journeys with a model to calculate emissions and 4) estimate the impact of carrying out these calculations at the European level on the quality of emissions calculations.

Task 4 – Future perspectives (envisaged under SGA-2)

Aim of this task is to summarise the project results and perform a qualitative cost-benefit analysis of using AIS-data for official statistics. These analyses should include aspects like sustainability of the data source, possibilities of improving international comparability, possibilities of data sharing (at micro- or aggregated level), quality improvement of current statistics and a sketch of a possible statistical process and needed infrastructure.

Deliverables (SGA-1 only):

4.1	Report on creating a database with AIS-data for official statistics: possibilities and pitfalls	Month 6
4.2	Report about deriving harbour visits and linking data from port authorities with AIS-data	Month 12
4.3	Report about sea traffic analyses using AIS-data	Month 18

Milestones (SGA-1 only):

4.4	Progress and technical report of first internal WP-meeting	Month 4
4.5	Progress and technical report of second internal WP-meeting	Month 9

Annex 2: SGA-2 of WP4 in more detail (original)

Work package number	4	Start date: End date:			1.8.2017 31.5.2018
Title	**AIS Data**				
Partner/co-beneficiary **(person days)**	**NL** **88**	DK 53	EL 66	NO 21	PL 35

Description of the work package

Aim of this work package is to investigate whether real-time measurement data of ship positions (measured by the so-called AIS-system) can be used 1) to improve the quality and internal comparability of existing statistics and 2) for new statistical products relevant for the ESS. Improvement of quality and internal comparability can be obtained e.g. by developing a reference frame of ships and their travels in European waters and then linking this reference frame, by ship number, to register-based data about marine transport from port authorities. These linked data can then be used for emission calculations. New products can be developed for e.g. traffic analyses. The added value of running a pilot with AIS-data at European level is that the source data are generic worldwide and data can be obtained at European level.

Challenges ahead with this dataset are: obtaining the data at European level, processing and collecting the data in such way that they can be used for multiple purposes, and visualising the results. A part of this work package is also to look into AIS analyses done by others and to investigate the possibility of obtaining already processed data as input for creating comparable official statistics. Especially it is important to make contact with other public authorities. This work package may require data acquisition in collaboration with Eurostat.

Methodological, quality and technical results of the work package, including intermediate findings, will be used as inputs for WP 8 of SGA-2. When carrying out the tasks listed below, care will be taken that these results will be stored for later use, by using the facilities described at WP 9.

SGA-2 of WP 4 will deliver the products below:

Task 3 – Methodology and Techniques

Estimate emissions

This task involves following individual vessels through time. Consequently we can infer the journeys from the data. Combined with a model to estimate the emission of vessels (which depends on travel distance, speed, draught, weather conditions and characteristics of the vessel itself), emissions of e.g. CO_2 and NO_x can be estimated per ship and per national territory.

Estimation of emissions based on only AIS data is impossible. At least, we need to know what the emission should be given, a.o., draught, speed and weather conditions. There are several sources that can be used to get information about the emission of vessels. For example Lloyds Register of Shipping.

Also, all ships have to register their emissions, so there is data available at the freight ship companies. A dataset for the same period as the AIS dataset could be used to model the emissions based on the variables mentioned above. We assume getting these needed data for free for this pilot.

Furthermore, satellite data is available for emissions (see, for instance http://www.globemission.eu), which shows very clearly the maritime routes. This satellite data could be used as a more direct source for measuring emissions and maybe for testing our model. Another possibility to test the model is to compare the estimated emissions by the model with the real fuel purchase or information coming from vessel's oil record book.

An advantage of doing these analyses on a European scale– instead of the national level – is that more precise estimates for emissions at national territories can be made.

Aim of this task is to develop and test a methodology for estimating vessel emissions based on AIS data by:

1) inferring journeys from AIS-data
2) visualising the results
3) investigating methodology for calculating emissions
4) combining these journeys with a model to calculate emissions
5) Testing the model and
6) estimating the impact of carrying out these calculations at the European level on the quality of emissions calculations.

Participants: Norway, Netherlands , Denmark, Greece, Poland

Task 4 – Access to and analysing AIS data from EMSA

AIS-data are available for national territories and the entire European territory. In this work package we use European AIS data from Dirkzwager. But for the future we would like to get free European AIS data. That is why we try to get AIS data at the European level from the European Maritime Safety Agency (EMSA).

In SGA-2 we focus on getting the European AIS data from EMSA and compare these two sources (AIS data from EMSA and Dirkzwager) on their coverage and quality. We will assess the quality of both sources by applying the quality framework. We also describe the strengths and weaknesses of both sources compared with existing maritime data.

This work package may require data acquisition in collaboration with Eurostat, because we would like Eurostat to apply for the data at EMSA's. Eurostat already received AIS data from EMSA for other projects and Eurostat can apply EMSA data for all the partner countries once. This will be better than applying for each country separated. If we do not get access to the EMSA data we cannot process this task and we cannot deliver deliverable 4.7, but it has not any consequences for the other deliverables in this WP. Not getting AIS data from EMSA means that we cannot get AIS data on the European level for free.

Participants: Netherlands , Denmark, Greece, Poland

Task 5 – New statistical output

Aim of this task is to explore possibilities of new statistical products (for example inter port statistics) by using AIS data. This task also includes analysing and elaborating scenarios for production of European and national statistics based on one single European data source of AIS data.

Participants: Norway, Netherlands, Denmark, Greece, Poland

Task 6 – Future perspectives

Aim of this task is to produce a consolidated report summarising the contents and the outcomes of WP 4. This report also includes a cost-benefit analysis of using AIS-data for official statistics. The report should also include aspects like sustainability of the data source, possibilities of improving international comparability, possibilities of data sharing (at micro- or aggregated level), possibilities of meeting the needs of both European and national statistics by one European AIS database, quality improvement of current statistics and a sketch of a possible statistical process and needed infrastructure**,** including technical skills required to generate statistical outputs from the source data.

Participants: Norway, Netherlands, Denmark, Greece, Poland

Deliverables (SGA-2 only):

4.6	Report on determining emissions. This report will describe the investigated methodology for calculating emissions and the reason why we choose for a certain methodology. The report also describes the created model itself (inclusive other needed data sources) and the results of testing the model. Finally this report also describes the impact of carrying out these calculations at the European level on the quality of emissions calculations.	Month 13
4.7	Report about the results of comparing the quality and coverage of the European AIS data from Dirkzwager and EMSA (by applying the quality framework). Also the strengths and weaknesses of both sources will be compared with existing maritime data.	Month 15
4.8	Report about possible new statistical output based on European AIS data. The report also describes analysed and elaborated scenarios for production of European and national statistics based on one single European data source of AIS data.	Month16
4.9	Consolidated report on project results including a cost-benefit analysis of using AIS-data for official statistics.	Month 17

Milestones (SGA-2 only):

4.10	Progress and technical report of first internal WP-meeting	Month 9
4.11	Progress and technical report of the second internal WP-meeting	Month 14

Annex 3: Adjusted scope SGA-2 of WP4 in more detail

Work package number	4	Start date: End date:			1.8.2017 31.5.2018
Title	**AIS Data**				
Partner/co-beneficiary **(person days)**	**NL** **88**	DK 53	EL 66	NO 21	PL 35

Description of the work package

Aim of this work package is to investigate whether real-time measurement data of ship positions (measured by the so-called AIS-system) can be used 1) to improve the quality and internal comparability of existing statistics and 2) for new statistical products relevant for the ESS. Improvement of quality and internal comparability can be obtained e.g. by developing a reference frame of ships and their travels in European waters and then linking this reference frame, by ship number, to register-based data about marine transport from port authorities. These linked data can then be used for emission calculations. New products can be developed for e.g. traffic analyses. The added value of running a pilot with AIS-data at European level is that the source data are generic worldwide and data can be obtained at European level.

Challenges ahead with this dataset are: obtaining the data at European level, processing and collecting the data in such way that they can be used for multiple purposes, and visualising the results. A part of this work package is also to look into AIS analyses done by others and to investigate the possibility of obtaining already processed data as input for creating comparable official statistics. Especially it is important to make contact with other public authorities. This work package may require data acquisition in collaboration with Eurostat.

Methodological, quality and technical results of the work package, including intermediate findings, will be used as inputs for WP 8 of SGA-2. When carrying out the tasks listed below, care will be taken that these results will be stored for later use, by using the facilities described at WP 9.

SGA-2 of WP 4 will deliver the products below:

Task 3 – Methodology and Techniques

Estimate emissions

This task involves following individual vessels through time. Consequently we can infer the journeys from the data. Combined with a model to estimate the emission of vessels (which depends on travel distance, speed, draught, weather conditions and characteristics of the vessel itself), emissions of e.g. CO_2 and NO_x can be estimated per ship and per national territory.

Estimation of emissions based on only AIS data is impossible. At least, we need to know what the emission should be given for example draught, speed and weather conditions. There are several sources that can be used to get information about the emission of vessels, such as Lloyds Register of Shipping.

Also, all ships have to register their emissions, so there is data available at the freight ship companies. A dataset for the same period as the AIS dataset could be used to model the emissions based on the variables mentioned above. We assume getting these needed data for free for this pilot.

Furthermore, satellite data is available for emissions (see, for instance http://www.globemission.eu), which shows very clearly the maritime routes. This satellite data could be used as a more direct source for measuring emissions and maybe for testing our model. Another possibility to test the model is to compare the estimated emissions by the model with the real fuel purchase or information coming from vessel's oil record book.

An advantage of doing these analyses on a European scale– instead of the national level – is that more precise estimates for emissions at national territories can be made.

Aim of this task is to investigate existing models for deriving emissions and describe how we could implement these models in our statistical processes. We also will investigate possibilities for improving emission statistics by using AIS data.

Participants: Norway, Netherlands , , Greece, Poland

Task 4 – Access to and analysing AIS data from EMSA

AIS-data are available for national territories and the entire European territory. In this work package we use European AIS data from Dirkzwager. But for the future we would like to get free European AIS data. That is why we (together with Eurostat) try to get AIS data at the European level from the European Maritime Safety Agency (EMSA).

In SGA-2 we focused on getting the European AIS data from EMSA. Unfortunately, we did not succeed in getting access to data from EMSA During SGA-2. We succeeded to get satellite data from LuxSpace available from the Mediterranean Sea for the same period as the Dirkzwager data. So we could assess the quality of satellite data by comparing it to the DZ data and the national AIS data from Greece.

Participants: Netherlands, Greece, Poland

Task 5 – New statistical output

Aim of this task is to explore possibilities of new statistical products (for example interport statistics) by using AIS data. This task also includes analysing and elaborating scenarios for production of European and national statistics based on one single European data source of AIS data.

Participants: Norway, Netherlands, Greece, Poland

Task 6 – Future perspectives

Aim of this task is to produce a consolidated report summarising the contents and the outcomes of WP 4. This report also includes a cost-benefit analysis of using AIS-data for official statistics. The report should also include aspects like sustainability of the data source, possibilities of improving international comparability, possibilities of data sharing (at micro- or aggregated level), possibilities of meeting the

needs of both European and national statistics by one European AIS database, quality improvement of current statistics and a sketch of a possible statistical process and needed infrastructure, including technical skills required to generate statistical outputs from the source data.

Participants: Norway, Netherlands, Greece, Poland

Deliverables (SGA-2 only):

4.6	Report on estimating emissions. This report will describe the results of investigating existing models for calculating emissions. The report also will describe the results of investigating possibilities for improving emission statistics by using AIS.	Month 13
4.3	Adjusting deliverable 4.3 on the quality of AIS data by with a comparison of the quality and coverage of the European AIS data from Dirkzwager and the Satellite data from LuxSpace. The satellite data will also be compared to the National AIS data from Greece.	Month 15
4.7	Report about possible new statistical output based on European AIS data. The report also describes analysed and elaborated scenarios for production of European and national statistics based on one single European data source of AIS data.	Month 16
4.8	Consolidated report on project results including a cost-benefit analysis of using AIS-data for official statistics.	Month 17

Milestones (SGA-2 only):

4.10	Progress and technical report of first internal WP-meeting	Month 9
4.11	Progress and technical report of the second internal WP-meeting	Month 14

Annex 4: IHS classification of ships used in the STEAM model

LNG Tanker	Container/Ro-Ro Cargo Ship	Pile Driving Vessel
LPG Tanker	Landing Craft	Icebreaker
LPG/Chemical Tanker	Passenger/Ro-Ro Ship (Vehicles)	Icebreaker/Research
CO2 Tanker	Passenger/Ro-Ro Ship (Vehicles/Rail)	Cable Repair Ship
Molten Sulphur Tanker	Passenger/Landing Craft	Cable Layer
Chemical Tanker	Passenger/Cruise	Incinerator
Chemical/Products Tanker	Passenger Ship	Waste Disposal Vessel
Wine Tanker	Livestock Carrier	Effluent carrier
Vegetable Oil Tanker	Barge Carrier	Fire Fighting Vessel
Edible Oil Tanker	Heavy Load Carrier	Pollution Control Vessel
Beer Tanker	Heavy Load Carrier, semi submersible	Patrol Vessel
Latex Tanker	Yacht Carrier, semi submersible	Crew Boat
Shuttle Tanker	Nuclear Fuel Carrier	Training Ship
Crude Oil Tanker	Nuclear Fuel Carrier (with Ro-Ro facility)	Utility Vessel
Crude/Oil Products Tanker	Pulp Carrier	Search & Rescue Vessel
Products Tanker	Factory Stern Trawler	Pilot Vessel
Tanker (unspecified)	Stern Trawler	Salvage Ship
Asphalt/Bitumen Tanker	Trawler	Buoy Tender
Coal/Oil Mixture Tanker	Fishing Vessel	Buoy & Lighthouse Tender
Water Tanker	Fish Factory Ship	Lighthouse Tender
Molasses Tanker	Fish Carrier	Supply Tender
Glue Tanker	Live Fish Carrier (Well Boat)	Mooring Vessel
Alcohol Tanker	Fish Farm Support Vessel	Work/Repair Vessel
Caprolactam Tanker	Fishery Patrol Vessel	Hospital Vessel
Fruit Juice Tanker	Fishery Research Vessel	Tank Cleaning Vessel
Bulk Carrier	Fishery Support Vessel	Trans Shipment Vessel
Bulk Carrier, Laker Only	Seal Catcher	Anchor handling Vessel
Bulk Carrier (with Vehicle Decks)	Whale Catcher	Log Tipping Ship
Ore Carrier	Kelp Dredger	Bunkering Tanker
Bulk/Oil Carrier (OBO)	Pearl Shells Carrier	Exhibition Vessel
Ore/Oil Carrier	Crew/Supply Vessel	Theatre Vessel
Bulk Cargo Carrier, self discharging	Pipe Carrier	Mission Ship
Bulk Cargo Carrier, self discharging, Laker	Platform Supply Ship	Bulk Dry Storage Ship
Cement Carrier	Anchor Handling Tug Supply	Bulk Cement Storage Ship
Wood Chips Carrier, self unloading	Offshore Tug/Supply Ship	Mining Vessel
Urea Carrier	Offshore Support Vessel	Power Station Vessel
Aggregates Carrier	Diving Support Vessel	Vessel (function unknown)
Limestone Carrier	Accommodation Ship	Sailing Vessel
Refined Sugar Carrier	Drilling Ship	Hopper/Dredger (unspecified)
Powder Carrier	Pipe Layer Crane Vessel	Hopper, Motor
General Cargo Ship (with Ro-Ro facility)	Pipe Layer	Stone Carrier
General Cargo, Self-discharging	Production Testing Vessel	Crane Ship
Open Hatch cargo Ship	FPSO, Oil	Suction Dredger
General Cargo/Tanker (Container/oil/bulk - COB ship)	Gas Processing Vessel	Dredger (unspecified)
General Cargo/Tanker	Well Stimulation Vessel	Bucket Hopper Dredger
General Cargo Ship	Standby Safety Vessel	Grab Hopper Dredger
Palletised Cargo Ship	FSO, Oil	Suction Hopper Dredger
Deck Cargo Ship	Trenching Support Vessel	Trailing Suction Hopper Dredger
General Cargo/Passenger Ship	Pipe Burying Vessel	Grab Dredger
Container Ship (Fully Cellular)	Research Survey Vessel	Backhoe Dredger
Container Ship (Fully Cellular with Ro-Ro Facility)	Tug	Bucket Wheel Suction Dredger
Passenger/Container Ship	Articulated Pusher Tug	Bucket Ladder Dredger
Refrigerated Cargo Ship	Pusher Tug	Cutter Suction Dredger
Ro-Ro Cargo Ship	Rail Vehicles Carrier	Vehicles Carrier